BC7215 Chip and Driver Library Troubleshooting

1. bc7215_data_ready() cannot detect data packets

This situation is usually caused by setting the `BC7215_MAX_RX_DATA_SIZE` parameter in the configuration file too small, so that the actual raw data length exceeds this setting. In general, data sent by remote controls for audio/video devices is usually within 8 bytes, while data from air-conditioner remote controls is usually from a dozen bytes to several dozen bytes.

Another possibility is that the data format transmitted by the remote control is not supported. The BC7215 does not support every encoding and modulation format; a very small number of formats may fall outside its supported range. To verify whether the issue is caused by the format, test with a more common type of remote control.

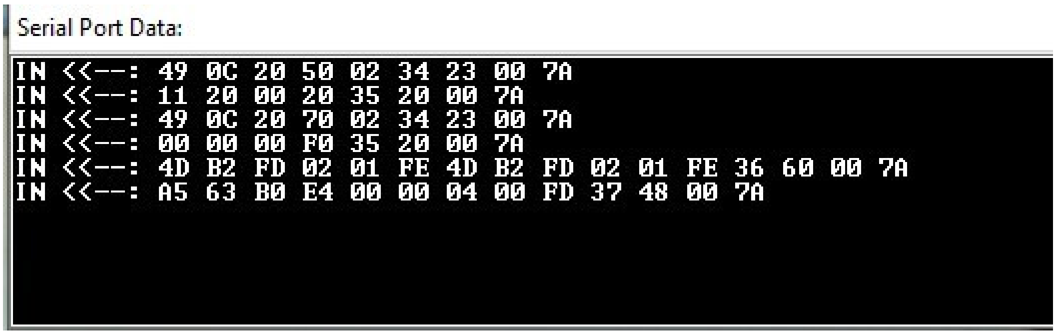
This problem may also be caused by hardware. If the infrared receiver module has poor performance or does not match the infrared signal being received, the signal input to the BC7215 may become distorted or lost. Some remote controls use very narrow pulses, as short as about 200 us. Some infrared receiver modules are suitable only for receiving wide-pulse signals and may have a minimum receivable pulse width of 400 us; in this case, the BC7215 may not be able to decode the signal correctly. In addition to the shortest pulse width, the carrier frequency and wavelength of the infrared receiver module may also have an effect. Remote-control carrier frequencies are mainly divided into 38 kHz and 56 kHz types, and infrared wavelengths include 850 nm and 940 nm types. Among these, 38 kHz and 940 nm are the most commonly used parameters.

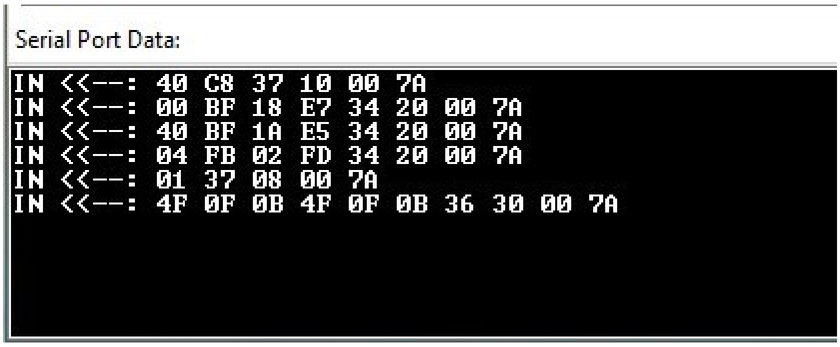
2. The BC7215 can receive data packets in Simple Mode, but only format packets in Complex Mode

This situation is also caused by setting the `BC7215_MAX_RX_DATA_SIZE` parameter too small. The receive buffer of the driver library is sized as the sum of the raw data packet and the format information packet. If the configured maximum processable length is smaller than the actual received data length, the packet may be received normally in Simple Mode because the entire buffer can be used for the data packet. However, in Complex Mode, the format information packet immediately follows the raw data packet. When the buffer is not large enough, the format information packet overwrites the raw data packet, so only the format information packet can be received.

3. Failure to learn infrared remote-control signals

There are many "universal" remote controls on the market. A single remote control may be able to control products of different brands and models. Some original factory remote controls can also control several different products from the same manufacturer. This type of remote control may transmit several different formats of remote-control signals continuously when the user presses one key. The two examples in the original document show the signals received when one key is pressed on a universal TV remote control and a universal air-conditioner remote control. Each remote control transmitted five signals with different formats and contents. If the learned signal is only one of these signals, it may not be the format corresponding to your device.





In this case, use the BC7215 demo software to record both the data and format information of each signal, and then test them one by one to determine which one corresponds to your device.

4. Unable to transmit infrared signals

This problem can be divided into several cases: no transmission at all, incorrect transmitted data, or the system hanging when a transmit function is called. To determine whether an infrared signal is being transmitted, observe the `OUT` pin of the BC7215 with an oscilloscope, or add an LED indicator to this pin for easier observation.

I. If no infrared signal is transmitted at all, check the following four items

1. Check whether the BC7215 is in transmit mode and whether it has entered shutdown mode.
2. Check whether the format information packet is correct. The BC7215 must have format information internally before it can transmit an infrared signal correctly. If no format information has been downloaded, or if no valid infrared signal was received before switching to transmit mode, correct transmission is definitely not possible. When switching directly to transmit mode after receiving an infrared signal, make sure that no other infrared signal is received between the time the infrared data is received and the time the device is switched to transmit mode. If the BC7215 receives new infrared pulses before it switches to transmit mode, it updates its internal format information. As a result, when it enters transmit mode, the internal format information has already changed. The safest method is to reload the format information before transmitting.
3. Check whether the loaded format information is complete. The BC7215 requires the loaded format information to exactly match the data output during decoding at that time. Errors in this information may cause the BC7215 to be completely unable to transmit.

4. Confirm that the UART parameters are correct. The parameters required by the BC7215 are: baud rate 19200, 8 data bits, 2 stop bits, and no parity. If these parameters are incorrect, the BC7215 cannot operate normally.

II. There is infrared output, but the receiving device cannot receive or recognize it

If infrared output is confirmed but the receiving device cannot receive or recognize it, check the following items:

1. **Hardware circuit**. Unlike an ordinary LED, the peak drive current of an infrared LED may need to reach tens or even hundreds of milliamps to achieve the expected receiving distance. Refer to the datasheet of the infrared LED being used. When the BC7215 directly drives the infrared LED, the receiving distance is relatively short. In general, it is recommended to add an external transistor driver.
2. **Carrier settings**. The BC7215 supports disabling the carrier and selecting between two different carrier frequencies. Carrier-free output is intended only for cases where the user uses a separate carrier generation circuit. If the selected carrier frequency is different from what the receiving device requires, the signal also cannot be received.
3. **Completeness of the infrared transmit command**. The `F5 02` command requires the data length that follows to be consistent with the actual amount of data sent over the UART. If the data length parameter to be transmitted does not match the actual UART data, especially when the actual data is shorter than the expected value, the infrared signal will be interrupted and the receiver will not be able to receive it correctly.
4. **Correctness of the data content**. The BC7215 supports transmitting arbitrary data, but if the receiving device is an audio/video product or similar device, it may impose certain requirements on the data content. For example, devices from a specific manufacturer may accept only data with a specific address code. The RC5 format requires one bit in the data to toggle each time, and the NEC format requires the last two bytes to be bitwise complements of each other. In this case, another BC7215 can be used to receive the data and verify whether the transmitted data is correct.
5. **Transmission interval**. Various infrared receiving devices, including the BC7215, distinguish different data frames by the time interval between them. The BC7215 chip does not automatically generate this interval during transmission. If continuous transmission is required, the user must ensure the time interval between two transmissions. A recommended interval is at least 40 ms.

III. The system hangs as soon as a transmit-related function is called

All infrared transmit-related functions involve sending data to the BC7215. When hardware UART flow control is not used, the driver library checks the state of the `BUSY` line before sending data. If it finds that `BUSY` is high, it waits indefinitely. If, for either hardware or software reasons, the driver library cannot read a low level on `BUSY` - the `BUSY` state is read through the user-provided function configured during initialization by `bc7215_config_read_busy_funct()` - the system will hang.

IV. Transmission is very unstable, and the success rate is very low

Check the UART settings. The BC7215 requires two stop bits. Most devices default to one stop bit. If the UART is not configured for two stop bits, the BC7215 may receive data unstably or incorrectly, causing it to fail to execute the commands expected by the user.